

FIBER-GRAND Paper I

Independent Claim-by-Claim Novelty Review

Exact aggregate-ML decoding for one uniform deletion and independent BSC substitutions

Review date	3 August 2026
Bundle reviewed	FIBER_GRAND_Paper_I_External_Review_Bundle_Phase2D_R1.zip
Integrity	SHA-256 verified: a485bbf4f655f73b4250f9048a7b913b07db8d25000f6d8737f06373c0d02cd0
Primary focus	Conference manuscript; claim novelty matrix; results checklist; reviewer handoff
Review type	Academic novelty and venue-positioning assessment; not a patent or legal opinion

Venue-level recommendation

ITW_READY_AFTER_REPAIR

The mathematical core appears publishable, but the present novelty framing overstates the originality of the discovery metric, aggregate-path inference, threshold logic, and query exponent.

1. Executive decision

ITW_READY_AFTER_REPAIR

The paper should not be submitted in its current framing as an ISIT-ready novelty package. It does contain a defensible narrow contribution suitable for an Information Theory Workshop paper after targeted repair. The strongest apparently distinct results are: (i) the explicit binary strict best-history versus aggregate-ML reversal family and its exact threshold; (ii) the channel-specific unseen-shell likelihood bound; and (iii) the realization-dependent stopping theorem $S \leq \min\{m, T + L_n(p_s) - 1\}$.

The broader decoder architecture is a synthesis of established ingredients: path-marginalized synchronization inference, inverse-noise membership search, minimum deletion-alignment distance, threshold or branch-and-bound certification, and classical Hamming-shell counting. The high-probability $h_2(p_s)$ query exponent is valuable as a consequence of the stopping theorem, but its entropy-volume mechanism is substantially anticipated by GRAND and typical-set search theory.

Most important novelty correction

Gershon and Cassuto (ISIT 2022) already define the minimum substitution distance over all single-deletion alignments and state linear computational complexity. This is the same semantic object as the manuscript discovery shell $h_y(x) = \min_j d_j(x, y)$, up to orientation. Proposition 2 is therefore not a new distance concept; at most, its first-appearance statement is an immediate inverse-enumeration consequence of a known metric.

Novelty concentration

- Core distinct claims: explicit strict reversal family; exact shell upper bound U_s ; realization-dependent stopping theorem.
- Partially anticipated claims: exact shell-certified decoder as a combination; all-alignment linear scoring recurrence; finite query bound; $h_2(p_s)$ exponent.
- Known components: aggregate path likelihood; minimum deletion-alignment distance; generic threshold stopping; membership-oracle interface; $n+1$ insertion-supersequence cardinality.
- Original but non-general evidence: the frozen exact finite-length campaign and its separated query, scoring, generation, and timing measurements.

Why the verdict is not stronger or weaker

Not ISIT_READY: the manuscript currently presents several known or direct-corollary ingredients as part of a four-contribution package, omits a direct antecedent for the discovery metric, and does not yet distinguish conceptual novelty from channel-specific specialization sharply enough.

Not MAJOR_REVISION or REJECT: no direct anticipation was located for the exact reversal threshold, the closed-form unseen-shell bound in this model, or the $T + L_n - 1$ stopping theorem. The required changes are principally claim narrowing, related-work repair, and evidence discipline; the channel model need not be broadened and the main proofs need not be replaced.

2. Scope, method, and classification standard

The review reconstructed each provisional contribution from the conference manuscript and the three audit files named in the request. It then compared the mathematical object, inference rule, stopping condition, and complexity conclusion against primary literature on GRAND, insertion/deletion/substitution decoding, hidden-path aggregation, exact most-probable-string search, threshold aggregation, deletion-channel ML, and synchronization search. Searches were current through 3 August 2026.

Label	Meaning used in this review
DISTINCT	No primary source reviewed was found to disclose the same technical claim. This is a bounded academic assessment, not proof of universal nonexistence.
PARTIALLY_ANTICIPATED	The exact specialization or formula differs, but the core mechanism, object, or asymptotic conclusion is already present.
KNOWN	The substantive claim is already disclosed; a new notation, proof phrasing, or implementation restatement does not restore novelty.
UNRESOLVED	The search did not support a reliable classification because the closest evidence was incomplete, inaccessible, or technically ambiguous.

The classification is claim-specific. For example, the generic fact that a most-probable path can differ from a most-probable aggregate string is KNOWN, while the manuscript's exact binary two-codeword construction and threshold can still be DISTINCT. Likewise, threshold stopping is KNOWN, while a newly derived channel-specific upper bound used by that stopping rule can be DISTINCT.

Search limitation

A novelty review can identify anticipation and materially narrow claims, but it cannot prove that no earlier source exists. Every DISTINCT label below should be read as "no direct anticipation located in the primary literature and targeted searches reviewed through the stated date."

3. Claim-by-claim novelty matrix

The matrix separates broad provisional contributions from their load-bearing subclaims. This avoids certifying an entire decoder as novel merely because one channel-specific formula is new.

ID	Provisional claim	Classification	Closest source	Technical comparison
C1	Aggregate candidate likelihood is the sum over compatible deletion alignments.	KNOWN	[2], [4]-[6]	Path-marginalized likelihoods for variable-length channels and deletion-channel ML are established. Equation (2) is the exact specialization to one uniform deletion plus BSC substitutions, not a new inference principle.
C2	All n alignment mismatch counts can be updated in $O(n)$ arithmetic.	PARTIALLY_ANTICIPATED	[7]	The same minimum one-deletion/substitution alignment distance is already computed in linear time. The manuscript recurrence gives all alignment scores, which is a useful explicit specialization but not a strong independent novelty claim.
C3	Explicit strict best-history/aggregate-ML reversal with threshold $n p_s > 1 + 2 p_s^2$.	DISTINCT	[6], [9], [10]	The general path-versus-string reversal is known, and deletion multiplicity in ML is known. No reviewed source gives this binary family or this exact threshold. Novelty is narrow: the construction, not the phenomenon.
C4	Discovery shell $h_y(x) = \min_j d_j(x,y)$, and first appearance at shell $h_y(x)$.	KNOWN	[7], [12]	The minimum Hamming mismatch over all single-deletion alignments is directly disclosed as the natural distance and is linear-time computable. First appearance under exhaustive inverse shell enumeration follows immediately from that definition.
C5	Shell-certified exact membership-oracle decoder for aggregate ML.	PARTIALLY_ANTICIPATED	[1]-[3], [8], [10], [11]	Membership-based inverse search, path aggregation, synchronization-state search, and incumbent-versus-bound termination are all established. The exact one-deletion+BSC assembly and its closed-form shell certificate are a new combination, not a new general decoding paradigm.
C6	Unseen candidate bound $U_s = p_s^{s+1}(1-p_s)^{(m-s-1)}$.	DISTINCT	[10], [11]	Generic potential-probability and threshold bounds are known, but no reviewed primary source gives this exact shell bound for aggregate one-deletion+BSC word likelihoods. The derivation is simple but channel-specific and useful.
C7	Strict incumbent $> U_s$ certifies the complete ML tie set.	PARTIALLY_ANTICIPATED	[10], [11]	Strict incumbent-versus-upper-bound termination is standard exact-search logic. The complete-tie-set consequence is correct and important, but its novelty comes only from the specific U_s bound, not from strict comparison itself.

ID	Provisional claim	Classification	Closest source	Technical comparison
C8	Realization-dependent stopping: $S \leq \min\{m, T + L_n(p_s) - 1\}$.	DISTINCT	[1]	GRAND and GRANDAB analyze guess counts and fixed abandonment thresholds, but GRANDAB is approximate. No reviewed source gives this exact code-independent synchronization-channel bound derived from aggregate alignment mass and an unseen-word certificate.
C9	Finite query bound $(n+1) \text{Vol}(m, S)$, with scoring no larger than membership calls.	PARTIALLY_ANTICIPATED	[1], [12]	The $n+1$ insertion sphere and Hamming-ball counting are classical. The bound is a direct accounting consequence of C8 and known combinatorics rather than an independent algorithmic principle.
C10	High-probability query exponent no larger than $h_2(p_s)$.	PARTIALLY_ANTICIPATED	[1]	Typical-set Hamming volume and Shannon-entropy query caps are established in GRANDAB. The exact synchronization decoder inherits the exponent because alignment multiplicity is polynomial and C8 controls the shell; the setting is new, the exponent mechanism is not.
C11	Exactly $n+1$ distinct binary one-insertion supersequences.	KNOWN	[12]	Classical insertion-sphere cardinality. It is an implementation device only.
C12	Generic threshold aggregation or incumbent-versus-frontier stopping.	KNOWN	[10], [11]	Threshold aggregation and exact potential-probability search already stop when an incumbent dominates every unseen object or prefix.
C13	Arbitrary code accessed only by a membership test.	KNOWN	[1]	This is a defining GRAND interface. The manuscript's distinction is that first hit is insufficient and discovered codewords require aggregate scoring.
C14	Frozen 996-trial exact finite-length evidence with separated operation counts.	DISTINCT	Bundle evidence	The specific campaign and measurements are original evidence. They do not establish universal latency, instance optimality, or general synchronization-channel performance.

No claim was assigned UNRESOLVED. This reflects sufficient evidence for classification, not a guarantee that the search universe is complete.

4. Closest prior-art map

The following primary sources are the load-bearing antecedents. Each anticipates a different layer of the manuscript, so no single paper makes the entire package KNOWN. The novelty question is therefore one of decomposition and combination.

Area	Closest primary source	What it anticipates
GRAND and query complexity	Duffy, Li, and Medard (2019) [1]	ML by likelihood-ordered noise guessing; codebook membership interface; guesswork complexity; approximate typical-set abandonment.
Exact IDS likelihood and search	Bahl and Jelinek (1975) [2]	Variable-length channel likelihood $\Pr\{Y X\}$; efficient computation; stack decoding for insertion, deletion, and substitution channels.
Synchronization-state decoding	Davey and MacKay (2001) [3]; Banerjee, Lenz, and Wachter-Zeh (2022) [8]	Drift-state trellises, probabilistic resynchronization, and sequential search for synchronization errors.
BSC plus deletion ML for codes	Yang et al. (2020) [4]	ML formulation for binary linear codes over BSCs with deletion-modeled synchronization errors; code-specific optimization and certification.
Single-deletion ML	Srinivasavaradhan et al. (2021) [5]; Sabary et al. (2020) [6]	Exact ML objectives for deletion channels; embedding or alignment multiplicity as the source of aggregate candidate probability.
Minimum deletion-alignment distance	Gershon and Cassuto (2022) [7]	Minimum Hamming distance over all single-deletion alignments, with linear computation; directly anticipates $h_y(x)$.
Path versus aggregate string	Graves et al. (2006) [9]; de la Higuera and Oncina (2013) [10]	Most probable path need not yield the most probable output string; exact search uses accumulated output probability and bounds on unfinished alternatives.
Threshold aggregation	Fagin, Lotem, and Naor (2003) [11]	Terminate when a threshold bounds all unseen objects below the current winners.
Insertion sphere	Levenshtein (1966) [12]	Classical single insertion/deletion combinatorics underlying the $n+1$ generator.

5. Detailed claim reviews

5.1 Aggregate one-deletion likelihood and linear scoring

Classification	Confidence	Closest primary source
KNOWN	High	Bahl and Jelinek (1975) [2]; Yang et al. (2020) [4]

Claim reviewed: for candidate x , exact ML uses $W(y|x) = (1/n) \sum_j p_s^{d_j(x,y)} (1-p_s)^{m-d_j(x,y)}$, rather than the probability of the single best deletion/substitution history.

Technical comparison: Bahl and Jelinek derive likelihoods for variable-length insertion/deletion/substitution channels and give an efficient procedure for $\Pr\{Y|X\}$. Later deletion-channel work makes the same aggregate principle explicit through embedding counts or alignment sums, and Yang et al. formulate ML decoding for binary linear codes under BSC noise with deletion-modeled synchronization errors. The manuscript has a particularly simple closed form because the deletion count is fixed at one and the deletion location is uniform, but that is a model specialization.

The $O(n)$ recurrence for all d_j values is useful exposition. Gershon and Cassuto already obtain the corresponding minimum over all deletion alignments in linear time. The exact recurrence for retaining every d_j was not located verbatim, so it is best described as a compact implementation lemma, not a leading novelty claim.

$$W(y|x) = (1/n) \sum_{j=1}^n p_s^{d_j(x,y)} (1-p_s)^{m-d_j(x,y)}$$

$$d_{\{j+1\}} = d_j - 1\{x_{\{j+1\}} \neq y_j\} + 1\{x_j \neq y_j\}$$

5.2 Explicit strict best-history versus aggregate-ML reversal

Classification	Confidence	Closest primary source
DISTINCT	Moderate-high	Graves et al. (2006) [9]; de la Higuera and Oncina (2013) [10]

Claim reviewed: for $y = 0^{\{n-3\}}10$, $x_A = 0^{\{n-3\}}101$, $x_B = 0^n$, and $C = \{x_A, x_B\}$, the most likely individual history favors x_A , while aggregate ML favors x_B if and only if $n p_s > 1 + 2 p_s^2$.

Technical comparison: weighted-path models have long distinguished the most probable path from the most probable output string. Graves et al. state that best-path decoding is not guaranteed to find the most probable label. De la Higuera and Oncina give an explicit probabilistic automaton example where the string with the most probable single parse is not the consensus string. Deletion-channel ML likewise weights candidates by the number or mass of compatible deletion histories. These sources anticipate the phenomenon and its explanation.

What remains distinct is the exact binary channel/code/observation family and the closed threshold $n p_s > 1 + 2 p_s^2$. Targeted searches did not locate this family, an equivalent parametric family, or the same threshold. The manuscript should therefore claim an explicit counterexample family, not discovery of the general path-versus-string gap.

Safe novelty wording

"We give an explicit two-codeword one-deletion+BSC family for which best-history and aggregate-word orderings strictly reverse, with an exact if-and-only-if threshold." Do not claim that the possibility of such a reversal is itself new.

5.3 Discovery shell and inverse-alignment first appearance

Classification	Confidence	Closest primary source
KNOWN	High	Gershon and Cassuto (2022) [7]

Claim reviewed: $h_y(x) = \min_j d_j(x, y)$, and candidate x first appears in the inverse-alignment enumeration at shell $h_y(x)$.

Technical comparison: Gershon and Cassuto define a shift-compensating distance for one deletion plus substitutions and prove that it equals the minimum Hamming distance over all possible deletion indices. They also emphasize its linear computational complexity. Up to which string is viewed as the longer supersequence, this is the same mathematical object as $h_y(x)$.

Once exhaustive substitution shells are paired with all one-insertion supersequences, "first appearance at the minimum alignment mismatch" follows directly from the enumeration definition. This is a correct proposition and useful bridge to the algorithm, but it should be presented as a known metric plus an immediate enumeration lemma. The current related-work list must add [7].

5.4 Shell-certified exact decoder as an architecture

Classification	Confidence	Closest primary source
PARTIALLY_ANTICIPATED	High	Duffy et al. (2019) [1]; de la Higuera and Oncina (2013) [10]

Claim reviewed: enumerate BSC shells, generate one-insertion supersequences, query code membership, evaluate the full alignment-summed likelihood only for newly discovered codewords, and stop when the incumbent dominates every unseen candidate.

Technical comparison: GRAND supplies the code-agnostic membership-oracle interface and likelihood-ordered inverse-noise search. Bahl-Jelinek and later synchronization decoders supply path-marginalized variable-length inference. De la Higuera and Oncina provide exact best-first search for the most probable string using an upper bound on uncompleted alternatives, and Fagin-Lotem-Naor provide the generic threshold principle. Gershon-Cassuto supplies the one-deletion/substitution minimum alignment distance. The manuscript assembles these ideas in a particularly clean exact one-deletion+BSC decoder.

The assembly is technically useful because ordinary GRAND first-hit logic fails after histories coalesce into candidate words. Nevertheless, the architecture should be classified as a distinct specialization or synthesis, not as a wholly new decoding paradigm. Its defensible novelty depends on C6 and C8 below.

5.5 Exact unseen-shell bound and complete tie-set certificate

Classification	Confidence	Closest primary source
DISTINCT	Moderate	de la Higuera and Oncina (2013) [10]; Fagin et al. (2003) [11]

Claim reviewed: after completing shell $s < m$, every unseen candidate obeys

$$W(y|x_{\text{unseen}}) \leq U_s$$

$$U_s = p_s^{s+1} (1-p_s)^{m-s-1}$$

Technical comparison: exact search with a potential-probability or frontier bound is known. Fagin-style threshold algorithms similarly bound every unseen object. What was not found in those sources is this closed-form upper bound for an aggregate word likelihood under exactly one uniform deletion and BSC substitutions.

The proof is elementary but nontrivial in its consequence: despite n compatible deletion alignments, an unseen word has every alignment at mismatch at least $s+1$, so the average of its n alignment terms remains bounded by the same largest term. This makes the certificate code-independent and avoids an extra factor n in U_s .

The strict comparison incumbent $> U_s$ is necessary to return the complete ML tie set. Strict incumbent-versus-bound logic is generic and should not be separately advertised as novel; the channel-specific U_s is the novel part.

5.6 Realization-dependent stopping theorem

Classification	Confidence	Closest primary source
DISTINCT	Moderate-high	Duffy et al. (2019) [1]

Claim reviewed: with T surviving-symbol substitutions, $\rho = p_s/(1-p_s)$, and $L_n(p_s) = \min\{\ell \geq 1: \rho^\ell < 1/n\}$, the exact decoder stops by

$$S \leq \min\{m, T + L_n(p_s) - 1\}.$$

Technical comparison: GRAND analyzes the race between the true noise rank and a false codeword hit, and GRANDAB truncates search at a fixed typical-set threshold. GRANDAB is deliberately approximate and can abandon without an ML answer. The manuscript theorem is different: it is a deterministic, realization-conditioned upper bound for an exact aggregate-ML decoder, independent of the code beyond nonemptiness and membership access.

The one-deletion aggregation creates a $1/n$ lower bound on the transmitted word likelihood from the true alignment; the unseen bound decreases by ρ per additional shell. Combining the two yields the logarithmic L_n penalty. No direct primary source with this theorem or an equivalent code-independent bound was located.

This is the strongest algorithmic novelty in the paper. It should be stated and proved prominently, with boundary cases $p_s = 0$ and $p_s = 1/2$ kept explicit and the exact complete-tie convention preserved.

5.7 Finite query bound and high-probability exponent

Classification	Confidence	Closest primary source
PARTIALLY_ANTICIPATED	High	Duffy et al. (2019) [1]; Levenshtein (1966) [12]

Claim reviewed: $Q_{\text{gen}}, Q_{\text{mem}} \leq (n+1) \text{Vol}(m, S)$, $Q_{\text{score}} \leq Q_{\text{mem}}$, and with probability at least $1 - n^{-c}$, $Q_{\text{mem}}, Q_{\text{score}} \leq 2^{\{n(h_2(p_s) + o(1))\}}$.

Technical comparison: the finite bound multiplies a classical Hamming-ball count by the classical $n+1$ one-insertion sphere. Its new input is only the stopping radius C8. The high-probability exponent follows from concentration of T and standard Hamming-volume asymptotics. GRANDAB already uses a Shannon-entropy typical-set query cap, whereas the expected optimal guesswork exponent is governed by a Renyi quantity. The manuscript correctly states a high-probability bound, not an expected-complexity identity.

The synchronization specialization is still useful: it proves that summing n alignments does not change the exponential query order because the alignment multiplicity and L_n penalty are subexponential. But $h_2(p_s)$ should be presented as a corollary of the exact stopping theorem plus standard counting, not as a separate foundational discovery.

Complexity wording that must remain

The factor $|C|/Q_{\text{score}}$ counts avoided complete candidate-likelihood evaluations. It is not a universal latency factor. Membership cost is code-dependent; generator attempts, distinct membership calls, complete scores, and wall-clock time must remain separate.

5.8 Classical generator, threshold principle, and membership interface

Classification	Confidence	Closest primary source
KNOWN	High	Levenshtein (1966) [12]; Fagin et al. (2003) [11]; Duffy et al. (2019) [1]

The $n+1$ distinct binary one-insertion-supersequence cardinality is classical. The generic rule of stopping when a current result dominates a bound on all unseen results is classical threshold or branch-and-bound logic. Accessing a code through a membership oracle is central to GRAND. These items should remain implementation context or contrast only.

The manuscript already disclaims ownership of these ingredients. That discipline should be strengthened by explicitly attaching each known component to its closest source and by avoiding contribution-list language that bundles them into the decoder as if they were jointly new.

5.9 Exact finite-length evidence

Classification	Confidence	Closest primary source
DISTINCT	High for the dataset; low for generalization	Frozen Phase-2C evidence in the reviewed bundle

The 996-trial campaign, exact integer scoring, agreement with exhaustive and completed branch-and-bound comparators, and separated accounting of membership calls versus complete likelihood evaluations are original evidence. They support implementation correctness and demonstrate that complete scoring can be far smaller than exhaustive codebook-wise scoring in the tested cells.

They do not support a universal speed claim. The evidence record itself requires the following caveats: only 16 $n=32$ exhaustive timing cases at $k=21$; branch-and-bound on a selected capped subset; faster performance in four of five $n=32$ branch-calibration cells but not universally; failure of the distinct generator to pass its wall-clock gate; no naturally sampled strict disjoint best-path versus aggregate-ML tie-set cases; and restriction to exactly one uniform deletion with hard BSC substitutions.

This contribution is therefore DISTINCT only as a specific reproducibility and validation record. It does not materially raise the conceptual novelty classification of C4, C11, C12, or C13.

6. Novelty dependency and scientific significance

The paper's novelty is not evenly distributed. Its logical chain is:

1. The strict reversal family proves that first-history decoding is not a safe extension of GRAND to the specified synchronization channel.
2. The known minimum-alignment shell metric makes inverse enumeration complete.
3. The distinct U_s bound turns the enumeration into an exact certified search over aggregate word likelihoods.
4. The distinct stopping theorem converts that certificate into a realization-dependent shell radius.
5. Known Hamming-volume asymptotics then give the partially anticipated $h_2(p_s)$ exponent.

Thus the paper has a coherent narrow theorem package even after removing known ingredients. Its principal scientific value is not a new universal synchronization decoder; it is an exact certificate-and-stopping analysis showing how a GRAND-style membership search can remain exact when multiple synchronization histories coalesce into the same candidate word.

Layer	Novelty status	Role in a repaired paper
Motivation theorem	DISTINCT	Lead with the explicit strict reversal and explain precisely what naive first-history decoding gets wrong.
Exact certificate	DISTINCT	Make U_s and complete-tie certification the central decoder theorem.
Stopping analysis	DISTINCT	Present $S \leq T + L_n - 1$ as the strongest complexity theorem.

7. Required repairs before submission

1. **Reframe the claimed novelty.** State that the decoder is a one-deletion+BSC specialization and synthesis. Claim ownership only of the explicit reversal family, the exact unseen-shell bound, the resulting exact certificate, and the realization-dependent stopping theorem.
2. **Add the missing direct antecedent.** Cite and compare Gershon and Cassuto (2022) for the minimum deletion-alignment substitution distance and linear computation. Recast Proposition 2 as a known metric plus an immediate inverse-enumeration property.
3. **Repair the path-aggregation literature.** Add Bahl-Jelinek for variable-length channel likelihood, Graves and de la Higuera-Oncina for path-versus-string inference, and Sabary or Srinivasavaradhan for deletion-channel ML aggregation.
4. **Repair the same-channel and synchronization-search literature.** Add Yang et al. for ML decoding of binary linear codes under BSC and deletion synchronization errors; keep Davey-MacKay and Banerjee-Lenz-Wachter-Zeh as trellis/sequential contrasts.
5. **Demote the exponent to a corollary.** Describe $h_2(p_s)$ as standard Hamming-volume asymptotics applied after the new exact stopping theorem. Explicitly distinguish high-probability query count from expected GRAND guesswork complexity.
6. **Preserve complete-tie exactness.** Retain the strict incumbent $> U_s$ condition and the $p_s=0$ and $p_s=1/2$ boundary discussion. Do not silently switch to a single arbitrary ML representative.
7. **Keep operation-count claims separated.** Maintain distinct metrics for generated hypotheses, distinct membership tests, complete likelihood scores, and wall-clock time. Do not translate $|C|/Q_{\text{score}}$ into universal latency.
8. **State every empirical limitation.** Include the 16-case exhaustive timing scope, selected capped branch-and-bound subset, four-of-five $n=32$ comparison, generator wall-clock failure, zero strict disjoint empirical tie sets, and exact channel restriction.

Single highest-value revision

Rewrite the title, abstract, and contribution paragraph around "exact shell certificates and stopping for aggregate ML" rather than around introduction of a broadly new synchronization-decoding architecture.

8. Recommended defensible contribution wording

A repaired contribution paragraph can use language of the following form:

For the channel with exactly one uniformly located deletion and independent BSC substitutions, we first give an explicit binary family in which the most likely compatible history and the aggregate-ML codeword strictly disagree, with an exact threshold. We then specialize inverse-noise membership search to candidate words whose likelihood sums over deletion alignments. The main technical results are a closed-form upper bound on every codeword not yet discovered by a completed substitution shell, an exact complete-tie stopping certificate, and a code-independent realization-dependent stopping bound. Standard Hamming-volume arguments then yield a high-probability query exponent $h_2(p_s)$. The decoder is evaluated with exact arithmetic while separately reporting generation, membership, complete scoring, and timing costs.

Claims that should not appear

- "The first exact decoder for synchronization errors." Exact likelihood and exact synchronization decoding have a long prior history.
- "A new deletion/substitution distance." The minimum over deletion-alignment Hamming distances is directly anticipated by [7].

- "A new threshold aggregation principle." The generic principle is known; only U_s is channel-specific.
- "A new h_2 complexity law for guessing decoding." The entropy-volume mechanism is standard; the new contribution is the exact stop theorem that permits it here.
- "Orders-of-magnitude universal speedup." The measured score-saving factor is not a universal wall-clock factor.

9. Venue-level recommendation

ITW_READY_AFTER_REPAIR

Rationale

The repaired paper has an appropriate ITW profile: a narrowly specified channel; one clean counterexample family; an exact search certificate; a realization-dependent stopping theorem; a direct entropy corollary; and exact finite-length validation. That is a coherent workshop contribution with clear mathematical and implementation value.

The current version is not ISIT_READY because its novelty package still depends too heavily on known components and direct corollaries, and because the missing Gershon-Cassuto comparison materially changes the status of the discovery-shell proposition. An ISIT resubmission could become plausible only after the repaired framing is tested against reviewers and, preferably, after adding a further theorem that is not a direct consequence of Hamming-volume counting - for example, a sharper expected or distributional query result, a lower bound showing near-optimality of the shell certificate, or a nontrivial extension that preserves exact aggregate certification.

The verdict is not MAJOR_REVISION because the needed repair fits within the existing model and theorem chain. It is not REJECT because C3, C6, and C8 remain defensible distinct claims after conservative prior-art decomposition.

10. Final status of the eight provisional contributions

Provisional contribution	Final classification	Scope note
Strict one-deletion+BSC reversal	DISTINCT	Exact family and threshold only; generic reversal is known.
Shell-certified exact decoder	PARTIALLY_ANTICIPATED	Distinct channel-specific certificate within a known search architecture.
Realization-dependent stop theorem	DISTINCT	No equivalent exact code-independent bound located.
Query exponent $h_2(p_s)$	PARTIALLY_ANTICIPATED	New setting and consequence; standard typical-volume mechanism.
$n+1$ insertion generator	KNOWN	Classical insertion sphere.
Threshold principle	KNOWN	Classical threshold and exact-search logic.
GRAND membership interface	KNOWN	Defining prior GRAND feature.
Finite-length evidence	DISTINCT	Original evidence record; limited generalization.

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Review boundary

This report assesses academic novelty and venue positioning for the exact channel and claims stated in the reviewed bundle. It does not assess patent validity, does not guarantee that every publication or unpublished manuscript has been located, and does not extend the conclusions to multiple deletions, insertions, soft information, transducer channels, hardware, or code design.